1.34 Matrices: Introduction

Worked Solutions

1 a)

2×2

1 b)

2×1

1 c)

1×3

1 d)

2×3

1 e)

3×3

1 f)

1×2

1 g)

3×1

1 h)

3×2

1 i)

5×6

2 a)

For example, $\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$.

2 b)

For example, $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$.

2 c)

For example, $\begin{pmatrix} a & b \\ c & d \\ e & f \\ g & h \end{pmatrix}$.

2 d)

For example, $\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$.

2 e)

For example, $\begin{pmatrix} a & b & c & d \end{pmatrix}$.

2 f)

For example, $\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \\ j & k & l \\ m & n & o \end{pmatrix}$.

2 g)

For example, $\begin{pmatrix} a & b & c & d & e \\ f & g & h & i & j \end{pmatrix}$.

3 a)

$$\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix} + \begin{pmatrix} 5 & 2 \\ -1 & 7 \end{pmatrix} = \begin{pmatrix} 7 & 5 \\ 0 & 9 \end{pmatrix}$$

3 b)

$$\begin{pmatrix} 6 & 2 \\ 3 & -1 \end{pmatrix} - \begin{pmatrix} 4 & -7 \\ 6 & 9 \end{pmatrix} = \begin{pmatrix} 2 & 9 \\ -3 & -10 \end{pmatrix}$$

3 c)

$$\begin{pmatrix} 8 & -3 & 1 \\ 2 & -9 & 3 \end{pmatrix} + \begin{pmatrix} 3 & 2 & -4 \\ -9 & -1 & 3 \end{pmatrix} = \begin{pmatrix} 11 & -1 & -3 \\ -7 & -10 & 6 \end{pmatrix}$$

3 d)

$$\begin{pmatrix} -3 & 10 \\ 2 & -6 \\ 9 & -2 \end{pmatrix} - \begin{pmatrix} 6 & -8 \\ -8 & -8 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} -9 & 18 \\ 10 & 2 \\ 10 & -4 \end{pmatrix}$$

4 a)

$$2\begin{pmatrix} 1 & -3 & -9 \\ 5 & 2 & 4 \end{pmatrix} = \begin{pmatrix} 2 & -6 & -18 \\ 10 & 4 & 8 \end{pmatrix}$$

4 b)

$$-5\begin{pmatrix} 3 & 2 & -1 \\ -6 & 5 & 2 \\ -2 & 3 & -12 \end{pmatrix} = \begin{pmatrix} -15 & -10 & 5 \\ 30 & -25 & -10 \\ 10 & -15 & 60 \end{pmatrix}$$

5 a)

$$\begin{pmatrix} 5 \\ 7 \end{pmatrix} + 2\begin{pmatrix} -3 \\ 4 \end{pmatrix} = \begin{pmatrix} -1 \\ 15 \end{pmatrix}$$

5 b)

$$3\begin{pmatrix} 3 & -2 \\ 1 & 4 \end{pmatrix} - \begin{pmatrix} 5 & 2 \\ 9 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -8 \\ -6 & 11 \end{pmatrix}$$

5 c)

$$\frac{3}{2}\begin{pmatrix} 8 & 4 \\ 20 & 3 \\ -3 & 2 \end{pmatrix} - 4\begin{pmatrix} -2 & 3 \\ -1 & 2 \\ 8 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 20 & -6 \\ 34 & -\frac{7}{2} \\ -\frac{73}{2} & 1 \end{pmatrix}$$

6 a)

$$\begin{aligned} \mathbf{A} + 2\mathbf{B} + 3\mathbf{C} &= \begin{pmatrix} 7 & 2 \\ -1 & 3 \end{pmatrix} + 2\begin{pmatrix} 2 & 0 \\ 1 & -2 \end{pmatrix} + 3\begin{pmatrix} 4 & 20 \\ 9 & -11 \end{pmatrix} \\ &= \begin{pmatrix} 7 & 2 \\ -1 & 3 \end{pmatrix} + \begin{pmatrix} 4 & 0 \\ 2 & -4 \end{pmatrix} + \begin{pmatrix} 12 & 60 \\ 27 & -33 \end{pmatrix} \\ &= \begin{pmatrix} 23 & 62 \\ 28 & -34 \end{pmatrix} \end{aligned}$$

6 b)

$$\mathbf{B} - \mathbf{A} - \mathbf{C} = \begin{pmatrix} 2 & 0 \\ 1 & -2 \end{pmatrix} - \begin{pmatrix} 7 & 2 \\ -1 & 3 \end{pmatrix} - \begin{pmatrix} 4 & 20 \\ 9 & -11 \end{pmatrix} = \begin{pmatrix} -9 & -22 \\ -7 & 6 \end{pmatrix}$$

6 c)

$$\begin{aligned} \mathbf{C} - 2(\mathbf{A} + \mathbf{B}) &= \begin{pmatrix} 4 & 20 \\ 9 & -11 \end{pmatrix} - 2\left(\begin{pmatrix} 7 & 2 \\ -1 & 3 \end{pmatrix} + \begin{pmatrix} 2 & 0 \\ 1 & -2 \end{pmatrix}\right) \\ &= \begin{pmatrix} 4 & 20 \\ 9 & -11 \end{pmatrix} - 2\begin{pmatrix} 9 & 2 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 20 \\ 9 & -11 \end{pmatrix} - \begin{pmatrix} 18 & 4 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} -14 & 16 \\ 9 & -13 \end{pmatrix} \end{aligned}$$

7)

$$\begin{pmatrix} a & 2 \\ 3 & b \end{pmatrix} + \begin{pmatrix} b & 4 \\ 2 & 2a \end{pmatrix} = \begin{pmatrix} 9 & 6 \\ 5 & 10 \end{pmatrix} \Rightarrow \begin{pmatrix} a+b & 6 \\ 5 & 2a+b \end{pmatrix} = \begin{pmatrix} 9 & 6 \\ 5 & 10 \end{pmatrix}$$

$$a + b = 9 \quad (1), \qquad 2a + b = 10 \quad (2)$$

$$\therefore a = 1 \text{ and } b = 8$$

8)

$$\begin{pmatrix} 4 & k \\ k & m \\ 6 & k \end{pmatrix} - \begin{pmatrix} 2m & 3k \\ m & 5k \\ 2 & m \end{pmatrix} = \begin{pmatrix} -4m & 6m \\ -4m & 16m \\ -2m & -4m \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 4-2m & -2k \\ k-m & m-5k \\ 4 & k-m \end{pmatrix} = \begin{pmatrix} -4m & 6m \\ -4m & 16m \\ -2m & -4m \end{pmatrix}$$

$$4 - 2m = -4m \Rightarrow 4 = -2m \Rightarrow m = -2$$

$$-2k = 6m \Rightarrow -2k = -12 \Rightarrow k = 6$$

9)

$$\begin{pmatrix} 2 & 5q \\ 8 & 3p \\ p & p \end{pmatrix} + q \begin{pmatrix} 2q & 4 \\ p & q \\ 3q & 2 \end{pmatrix} = \begin{pmatrix} 4 & -p \\ q & 28 \\ 12 & 7 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2+2q^2 & 9q \\ 8+pq & 3p+q^2 \\ p+3q^2 & p+2q \end{pmatrix} = \begin{pmatrix} 4 & -p \\ q & 28 \\ 12 & 7 \end{pmatrix}$$

$$9q = -p \Rightarrow p + 9q = 0 \quad (1)$$

$$p + 2q = 7 \quad (2)$$

$$\therefore p = 9 \text{ and } q = -1$$

10)

Matrices $\mathbf{A}$ and $\mathbf{B}$ have different dimensions (2×2 and 2×3 respectively), and so we cannot find $\mathbf{A} + \mathbf{B}$.

11 a)

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$$

$$\mathbf{B} + \mathbf{A} = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$$

As $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$, this shows that the matrices $\mathbf{A}$ and $\mathbf{B}$ are commutative under addition.

11 b)

$$\mathbf{A}-\mathbf{B}=\begin{pmatrix}1&1\\1&1\end{pmatrix}-\begin{pmatrix}3&3\\3&3\end{pmatrix}=\begin{pmatrix}-2&-2\\-2&-2\end{pmatrix}$$

$$\mathbf{B}-\mathbf{A}=\begin{pmatrix}3&3\\3&3\end{pmatrix}-\begin{pmatrix}1&1\\1&1\end{pmatrix}=\begin{pmatrix}2&2\\2&2\end{pmatrix}$$

As $\mathbf{A}-\mathbf{B}\neq\mathbf{B}-\mathbf{A}$, this shows that in general matrices are not commutative under subtraction.

12 a)

$$\mathbf{A}+(\mathbf{B}+\mathbf{C})=\begin{pmatrix}1&1\\1&1\end{pmatrix}+\left(\begin{pmatrix}3&3\\3&3\end{pmatrix}+\begin{pmatrix}5&5\\5&5\end{pmatrix}\right)=\begin{pmatrix}1&1\\1&1\end{pmatrix}+\begin{pmatrix}8&8\\8&8\end{pmatrix}=\begin{pmatrix}9&9\\9&9\end{pmatrix}$$

$$((\mathbf{A}+\mathbf{B})+\mathbf{C})=\left(\begin{pmatrix}1&1\\1&1\end{pmatrix}+\begin{pmatrix}3&3\\3&3\end{pmatrix}\right)+\begin{pmatrix}5&5\\5&5\end{pmatrix}=\begin{pmatrix}4&4\\4&4\end{pmatrix}+\begin{pmatrix}5&5\\5&5\end{pmatrix}=\begin{pmatrix}9&9\\9&9\end{pmatrix}$$

As $\mathbf{A}+(\mathbf{B}+\mathbf{C})=(\mathbf{A}+\mathbf{B})+\mathbf{C}$, this shows that the matrices $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are associative under addition.

12 b)

$$\mathbf{A}-(\mathbf{B}-\mathbf{C})=\begin{pmatrix}1&1\\1&1\end{pmatrix}-\left(\begin{pmatrix}3&3\\3&3\end{pmatrix}-\begin{pmatrix}5&5\\5&5\end{pmatrix}\right)=\begin{pmatrix}1&1\\1&1\end{pmatrix}-\begin{pmatrix}-2&-2\\-2&-2\end{pmatrix}=\begin{pmatrix}3&3\\3&3\end{pmatrix}$$

$$(\mathbf{A}-\mathbf{B})-\mathbf{C}=\left(\begin{pmatrix}1&1\\1&1\end{pmatrix}-\begin{pmatrix}3&3\\3&3\end{pmatrix}\right)-\begin{pmatrix}5&5\\5&5\end{pmatrix}=\begin{pmatrix}-2&-2\\-2&-2\end{pmatrix}-\begin{pmatrix}5&5\\5&5\end{pmatrix}=\begin{pmatrix}-7&-7\\-7&-7\end{pmatrix}$$

As $\mathbf{A}-(\mathbf{B}-\mathbf{C})\neq(\mathbf{A}-\mathbf{B})-\mathbf{C}$, this shows that in general matrices are not associative under subtraction.

13)

$$\mathbf{A}-\mathbf{B}=\begin{pmatrix}2(x+1)&3(1-x)\\-3(2-x)&4(5-x)\end{pmatrix}-\begin{pmatrix}8x+1&5x-2\\2-x&3+x\end{pmatrix}$$

$$=\begin{pmatrix}2x+2&3-3x\\-6+3x&20-4x\end{pmatrix}-\begin{pmatrix}8x+1&5x-2\\2-x&3+x\end{pmatrix}=\begin{pmatrix}1-6x&5-8x\\4x-8&17-5x\end{pmatrix}$$

14)

$$\mathbf{A}-\mathbf{B}=\begin{pmatrix}7k+3&3+k\\5k+1&9-6k\end{pmatrix}-\begin{pmatrix}4k-3&13+6k\\5+7k&1-10k\end{pmatrix}$$

$$=\begin{pmatrix}3k+6&-10-5k\\-2k-4&8+4k\end{pmatrix}=\begin{pmatrix}3(k+2)&-5(k+2)\\-2(k+2)&4(k+2)\end{pmatrix}=(k+2)\begin{pmatrix}3&-5\\-2&4\end{pmatrix}$$

So $f(k)=k+2$.

15 a)

$$\mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

15 b)

$$3\mathbf{I} = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$$

15 c)

$$\mathbf{0} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

16 a)

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

16 b)

$$-2\mathbf{I} = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

16 c)

$$\mathbf{0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

17 a)

$$\mathbf{0} - \mathbf{A} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 3 & -7 \\ -2 & -5 \end{pmatrix} = \begin{pmatrix} -3 & 7 \\ 2 & 5 \end{pmatrix}$$

17 b)

$$\mathbf{A} + \mathbf{I} = \begin{pmatrix} 3 & -7 \\ -2 & -5 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -7 \\ -2 & -4 \end{pmatrix}$$

17 c)

$$2\mathbf{I} - 3\mathbf{A} = 2\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - 3\begin{pmatrix} 3 & -7 \\ -2 & -5 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} 9 & -21 \\ -6 & -15 \end{pmatrix} = \begin{pmatrix} -7 & 21 \\ 6 & 17 \end{pmatrix}$$

17 d)

$$
\begin{aligned}
&\mathbf{I} - (2\mathbf{A} + \mathbf{0}) \\
&= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - 2\left(\begin{pmatrix} 3 & -7 \\ -2 & -5 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\right) \\
&= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - 2\begin{pmatrix} 3 & -7 \\ -2 & -5 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 6 & -14 \\ -4 & -10 \end{pmatrix} \\
&= \begin{pmatrix} -5 & 14 \\ 4 & 11 \end{pmatrix}
\end{aligned}
$$

17 e)

$$
\begin{aligned}
&\mathbf{0} - (\mathbf{I} - 3\mathbf{A}) \\
&= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} - \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - 3\begin{pmatrix} 3 & -7 \\ -2 & -5 \end{pmatrix}\right) \\
&= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} - \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 9 & -21 \\ -6 & -15 \end{pmatrix}\right) \\
&= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} -8 & 21 \\ 6 & 16 \end{pmatrix} \\
&= \begin{pmatrix} 8 & -21 \\ -6 & -16 \end{pmatrix}
\end{aligned}
$$

17 f)

$$
\begin{aligned}
&10\mathbf{A} - (6\mathbf{I} - 5\mathbf{A}) \\
&= 10\begin{pmatrix} 3 & -7 \\ -2 & -5 \end{pmatrix} - \left(6\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - 5\begin{pmatrix} 3 & -7 \\ -2 & -5 \end{pmatrix}\right) \\
&= \begin{pmatrix} 30 & -70 \\ -20 & -50 \end{pmatrix} - \left(\begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix} - \begin{pmatrix} 15 & -35 \\ -10 & -25 \end{pmatrix}\right) \\
&= \begin{pmatrix} 30 & -70 \\ -20 & -50 \end{pmatrix} - \begin{pmatrix} -9 & 35 \\ 10 & 31 \end{pmatrix} \\
&= \begin{pmatrix} 39 & -105 \\ -30 & -81 \end{pmatrix}
\end{aligned}
$$

18)

$$
4\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & 3 \end{pmatrix} - \begin{pmatrix} b & 0 & 0 \\ 0 & 2c & 0 \\ 0 & 0 & a \end{pmatrix} = 2\mathbf{I}
$$

$$\Rightarrow \begin{pmatrix} 4a & 0 & 0 \\ 0 & 4b & 0 \\ 0 & 0 & 12 \end{pmatrix} - \begin{pmatrix} b & 0 & 0 \\ 0 & 2c & 0 \\ 0 & 0 & a \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$4a - b = 2 \ (1) \qquad 4b - 2c = 2 \ (2) \qquad 12 - a = 2 \ (3)$$

Solve (3) to get $a = 10$.

Substitute into (1): $4(10) - b = 2 \Rightarrow b = 38$.

Substitute into (2): $4(38) - 2c = 2 \Rightarrow c = 75$.

$\therefore a = 10, b = 38, c = 75$.

19)

$$2\begin{pmatrix} p & q \\ 3 & r \end{pmatrix} + k\mathbf{I} = 4\begin{pmatrix} 2q & r \\ p & 2p \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2p & 2q \\ 6 & 2r \end{pmatrix} + \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} = \begin{pmatrix} 8q & 4r \\ 4p & 8p \end{pmatrix}$$

$$2p + k = 8q \ (1) \quad 2q = 4r \Rightarrow q - 2r = 0 \ (2) \quad 6 = 4p \Rightarrow p = \frac{3}{2} \ (3) \quad 2r + k = 8p \ (4)$$

Substitute (3) into (1) and (4): $2(\frac{3}{2}) + k = 8q \Rightarrow k = 8q - 3$ (5), and $2r + k = 8(\frac{3}{2}) \Rightarrow k = 12 - 2r$ (6).

Solving (5) and (6): $8q - 3 = 12 - 2r \Rightarrow 8q + 2r = 15$ (7).

Solve (2) and (7): $q = \frac{5}{3}$ and $r = \frac{5}{6}$.

$\therefore p = \frac{3}{2}, q = \frac{5}{3}$ and $r = \frac{5}{6}$.

20)

AB can be calculated as **A** has the same number of columns as the number of rows of **B** (**A** has 2 columns and **B** has 2 rows). **BA** cannot be calculated as **B** has 2 columns and **A** has 3 rows, so these do not match.

21)

AB can be calculated as **A** has the same number of columns as the number of rows of **B** (**A** has 3 columns and **B** has 3 rows). **BA** cannot be calculated as **B** has 4 columns and **A** has 3 rows, so these do not match.

22)

A is a square matrix and so $\mathbf{A}^2$ can be calculated (**A** has 3 columns and 3 rows, so these match). **B** is not a square matrix and so $\mathbf{B}^2$ cannot be calculated (**B** has 3 columns and 2 rows, so these do not match).

23 a)

$(2 \times 2) \times (2 \times 4) = 2 \times 4$

23 b)

$(2 \times 4) \times (3 \times 4)$ cannot be calculated

23 c)

$(3 \times 2) \times (2 \times 2) = 3 \times 2$

23 d)

$(3 \times 2) \times (3 \times 1)$ cannot be calculated

23 e)

$(3 \times 1) \times (3 \times 2)$ cannot be calculated

23 f)

$(4 \times 2) \times (2 \times 4) = 4 \times 4$

23 g)

$(3 \times 4) \times (4 \times 2) = 3 \times 2$

23 h)

$(2 \times 4) \times (2 \times 2)$ cannot be calculated

24 a)

$$\begin{pmatrix} 2 & 3 & -4 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = (2 \times 2 + 3 \times (-1) + (-4) \times 2) = (-7)$$

24 b)

$$\begin{pmatrix} -3 & 1 & -2 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ -4 \\ 3 \\ 1 \end{pmatrix} = ((-3) \times 5 + 1 \times (-4) + (-2) \times 3 + 4 \times 1) = (-21)$$

24 c)

$$\begin{pmatrix} 4 & 3 \\ -1 & 2 \\ 5 & -6 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \times 4 + 3 \times 2 \\ (-1) \times 4 + 2 \times 2 \\ 5 \times 4 + (-6) \times 2 \end{pmatrix} = \begin{pmatrix} 22 \\ 0 \\ 8 \end{pmatrix}$$

24 d)

$$\begin{pmatrix} 3 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 1 & 5 \end{pmatrix} = \begin{pmatrix} 3 \times 2 + 1 \times 1 & 3 \times 0 + 1 \times 5 \\ 0 \times 2 + 1 \times 1 & 0 \times 0 + 1 \times 5 \\ 1 \times 2 + 0 \times 1 & 1 \times 0 + 0 \times 5 \end{pmatrix} = \begin{pmatrix} 7 & 5 \\ 1 & 5 \\ 2 & 0 \end{pmatrix}$$

24 e)

$$\begin{pmatrix} 5 & 1 & 1 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 5\times 3+1\times 1+1\times 1 & 5\times 2+1\times 2+1\times 1 \\ 1\times 3+2\times 1+1\times 1 & 1\times 2+2\times 2+1\times 1 \end{pmatrix} = \begin{pmatrix} 17 & 13 \\ 6 & 7 \end{pmatrix}$$

24 f)

$$\begin{pmatrix} 3 & 4 & 1 \\ 2 & 3 & 0 \\ 0 & 3 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 6 \end{pmatrix} = \begin{pmatrix} 3\times 1+4\times 2+1\times 6 \\ 2\times 1+3\times 2+0\times 6 \\ 0\times 1+3\times 2+5\times 6 \end{pmatrix} = \begin{pmatrix} 17 \\ 8 \\ 36 \end{pmatrix}$$

24 g)

$$\begin{aligned} &(3 \quad 2 \quad -5) \begin{pmatrix} -1 & -2 \\ 2 & 6 \\ 9 & 0 \end{pmatrix} \\ &= (3\times -1+2\times 2-5\times 9 \quad 3\times -2+2\times 6-5\times 0) \\ &= (-44 \quad 6) \end{aligned}$$

24 h)

$$\begin{aligned} &\begin{pmatrix} 4 & 2 \\ -3 & -1 \end{pmatrix} \begin{pmatrix} 5 & 2 & -9 \\ -3 & 1 & 10 \end{pmatrix} \\ &= \begin{pmatrix} 4\times 5+2\times (-3) & 4\times 2+2\times 1 & 4\times (-9)+2\times 10 \\ -3\times 5-1\times (-3) & -3\times 2-1\times 1 & -3\times (-9)-1\times 10 \end{pmatrix} \\ &= \begin{pmatrix} 14 & 10 & -16 \\ -12 & -7 & 17 \end{pmatrix} \end{aligned}$$

24 i)

$$\begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix} \begin{pmatrix} 1 & -2 & 3 & -8 \end{pmatrix} = \begin{pmatrix} 4\times 1 & 4\times (-2) & 4\times 3 & 4\times (-8) \\ -5\times 1 & -5\times (-2) & -5\times 3 & -5\times (-8) \\ 2\times 1 & 2\times (-2) & 2\times 3 & 2\times (-8) \end{pmatrix} = \begin{pmatrix} 4 & -8 & 12 & -32 \\ -5 & 10 & -15 & 40 \\ 2 & -4 & 6 & -16 \end{pmatrix}$$

24 j)

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \begin{pmatrix} b & c & a \end{pmatrix} = \begin{pmatrix} ab & ac & a^2 \\ b^2 & bc & ab \\ bc & c^2 & ac \end{pmatrix}$$

25)

$$\mathbf{ABA} = \begin{pmatrix} 5 & 2 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix} \begin{pmatrix} 5 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 5 & 2 & 1 \end{pmatrix} \begin{pmatrix} 10 & 4 & 2 \\ -15 & -6 & -3 \\ 20 & 8 & 4 \end{pmatrix}$$
$$= \begin{pmatrix} 40 & 16 & 8 \end{pmatrix}$$

26 a)

$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 \times 2 + 0 \times 1 & 1 \times 2 + 0 \times 1 \\ 1 \times 2 + 1 \times 1 & 1 \times 2 + 1 \times 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 3 & 3 \end{pmatrix}$$

26 b)

$$\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + 0 \times 2 & 1 \times 0 + 0 \times 1 \\ 2 \times 1 + 1 \times 2 & 2 \times 0 + 1 \times 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}$$

26 c)

$$\begin{pmatrix} 3 & 1 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 3 & 2 \\ 1 & 5 \end{pmatrix} = \begin{pmatrix} 3 \times 3 + 1 \times 1 & 3 \times 2 + 1 \times 5 \\ 2 \times 3 + 5 \times 1 & 2 \times 2 + 5 \times 5 \end{pmatrix} = \begin{pmatrix} 10 & 11 \\ 11 & 29 \end{pmatrix}$$

26 d)

$$\begin{pmatrix} -1 & 2 \\ -3 & 6 \end{pmatrix} \begin{pmatrix} 1 & 5 \\ -4 & 2 \end{pmatrix} = \begin{pmatrix} -1 \times 1 + 2 \times (-4) & -1 \times 5 + 2 \times 2 \\ -3 \times 1 + 6 \times (-4) & -3 \times 5 + 6 \times 2 \end{pmatrix} = \begin{pmatrix} -9 & -1 \\ -27 & -3 \end{pmatrix}$$

26 e)

$$\begin{pmatrix} 8 & -3 \\ -4 & 2 \end{pmatrix} \begin{pmatrix} -6 & 5 \\ 3 & -5 \end{pmatrix} = \begin{pmatrix} 8 \times (-6) - 3 \times 3 & 8 \times 5 - 3 \times (-5) \\ -4 \times (-6) + 2 \times 3 & -4 \times 5 + 2 \times (-5) \end{pmatrix} = \begin{pmatrix} -57 & 55 \\ 30 & -30 \end{pmatrix}$$

26 f)

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}^3$$
$$= \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$
$$= \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$$
$$= \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix} \quad \text{Alternative Method: } \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}^3$$
$$= (2\mathbf{I})^3$$
$$= 2^3\mathbf{I}^3$$
$$= 8\mathbf{I}$$
$$= \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix}$$

26 g)

$$\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}^3 = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 4 & 4 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 8 & 12 \\ 0 & 8 \end{pmatrix}$$

26 h)

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$$
$$= \begin{pmatrix} \cos^2\theta - \sin^2\theta \\ \sin\theta\cos\theta + \sin\theta\cos\theta \end{pmatrix}$$
$$= \begin{pmatrix} \cos^2\theta - \sin^2\theta \\ 2\sin\theta\cos\theta \end{pmatrix}$$

This is equal to $\begin{pmatrix} \cos 2\theta \\ \sin 2\theta \end{pmatrix}$

using Double Angle Formulae met in A-Level Maths.

27)

$$\mathbf{A}^2 - \mathbf{B}^2 = \begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix}^2 - \begin{pmatrix} 2 & 3 \\ 2 & 5 \end{pmatrix}^2 = \begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 3 & 2 \end{pmatrix} - \begin{pmatrix} 2 & 3 \\ 2 & 5 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ 2 & 5 \end{pmatrix}$$
$$= \begin{pmatrix} 4 & 3 \\ 9 & 7 \end{pmatrix} - \begin{pmatrix} 10 & 21 \\ 14 & 31 \end{pmatrix} = \begin{pmatrix} -6 & -18 \\ -5 & -24 \end{pmatrix}$$

28)

$$(\mathbf{A}-\mathbf{I})^4 = \left(\begin{pmatrix}1 & -3\\ 8 & 1\end{pmatrix} - \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\right)^4 = \begin{pmatrix}0 & -3\\ 8 & 0\end{pmatrix}^4$$

$$= \begin{pmatrix}0 & -3\\ 8 & 0\end{pmatrix}\begin{pmatrix}0 & -3\\ 8 & 0\end{pmatrix}\begin{pmatrix}0 & -3\\ 8 & 0\end{pmatrix}\begin{pmatrix}0 & -3\\ 8 & 0\end{pmatrix} = \begin{pmatrix}-24 & 0\\ 0 & -24\end{pmatrix}\begin{pmatrix}-24 & 0\\ 0 & -24\end{pmatrix} = \begin{pmatrix}576 & 0\\ 0 & 576\end{pmatrix} = 576\mathbf{I}$$

So $k = 576$.

29)

$$\begin{pmatrix}k & 0 & 1\\ 1 & k & 2\end{pmatrix}\begin{pmatrix}k & 1\\ 2 & k\\ 1 & k\end{pmatrix} = \begin{pmatrix}1-2i & 2-2i\\ 5-3i & 3-4i\end{pmatrix}$$

$$\Rightarrow \begin{pmatrix}k\times k+0\times 2+1\times 1 & k\times 1+0\times k+1\times k\\ 1\times k+k\times 2+2\times 1 & 1\times 1+k\times k+2\times k\end{pmatrix} = \begin{pmatrix}1-2i & 2-2i\\ 5-3i & 3-4i\end{pmatrix}$$

$$\Rightarrow \begin{pmatrix}k^2+1 & 2k\\ 3k+2 & k^2+2k+1\end{pmatrix} = \begin{pmatrix}1-2i & 2-2i\\ 5-3i & 3-4i\end{pmatrix}$$

$$2k = 2-2i \Rightarrow k = 1-i$$

Check the other elements:

$$(1-i)^2+1 = (1-2i-1)+1 = 1-2i \text{ as required}$$

$$3(1-i)+2 = 3-3i+2 = 5-3i \text{ as required}$$

$$(1-i)^2+2(1-i)+1 = (1-2i-1)+2-2i+1 = 3-4i \text{ as required}$$

$$\therefore k = 1-i$$

30 a)

$$\mathbf{AB} = \begin{pmatrix}4 & -i\\ 3 & 2i\end{pmatrix}\begin{pmatrix}k & -3i\\ 4i & -2\end{pmatrix} = \begin{pmatrix}4k+4 & -10i\\ 3k-8 & -13i\end{pmatrix}$$

30 b)

$$\mathbf{BA} = \begin{pmatrix}k & -3i\\ 4i & -2\end{pmatrix}\begin{pmatrix}4 & -i\\ 3 & 2i\end{pmatrix} = \begin{pmatrix}4k-9i & 6-ki\\ -6+16i & 4-4i\end{pmatrix}$$

30 c)

If $\mathbf{AB} = \mathbf{BA}$ then $4k+4 = 4k-9i \Rightarrow 4 = -9i$, which is not true.

$\therefore$ there is no value of k such that $\mathbf{AB} = \mathbf{BA}$.

31)

$$\mathbf{AC} = \begin{pmatrix} 2 & -1 & 3 \\ 3 & -9 & 10 \end{pmatrix} \begin{pmatrix} -2 & 7 \\ 4 & -6 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -11 & 23 \\ -52 & 85 \end{pmatrix}$$

$$\mathbf{BA} = \begin{pmatrix} 5 & 6 \\ -9 & 20 \end{pmatrix} \begin{pmatrix} 2 & -1 & 3 \\ 3 & -9 & 10 \end{pmatrix} = \begin{pmatrix} 28 & -59 & 75 \\ 42 & -171 & 173 \end{pmatrix}$$

$$\mathbf{CA} = \begin{pmatrix} -2 & 7 \\ 4 & -6 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 & 3 \\ 3 & -9 & 10 \end{pmatrix} = \begin{pmatrix} 17 & -61 & 64 \\ -10 & 50 & -48 \\ 1 & -8 & 7 \end{pmatrix}$$

$$\mathbf{CB} = \begin{pmatrix} -2 & 7 \\ 4 & -6 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 5 & 6 \\ -9 & 20 \end{pmatrix} = \begin{pmatrix} -73 & 128 \\ 74 & -96 \\ -14 & 14 \end{pmatrix}$$

32 a)

$$\begin{pmatrix} a & 2 \\ 2a & 1 \end{pmatrix} \begin{pmatrix} 3 & a \\ a & 1 \end{pmatrix} = \begin{pmatrix} 3a+2a & a^2+2 \\ 6a+a & 2a^2+1 \end{pmatrix} = \begin{pmatrix} 5a & a^2+2 \\ 7a & 2a^2+1 \end{pmatrix}$$

32 b)

$$\begin{pmatrix} 2b & b \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 4 & 3b \\ 2 & 2b \end{pmatrix} = \begin{pmatrix} 8b+2b & 6b^2+2b^2 \\ 8 & 3b+4b \end{pmatrix} = \begin{pmatrix} 10b & 8b^2 \\ 8 & 7b \end{pmatrix}$$

33)

$$\begin{aligned} \mathbf{AB} &= \begin{pmatrix} -2 & k & 0 \\ 1 & k & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & k-1 \\ -1 & 2k \end{pmatrix} \\ &= \begin{pmatrix} -2+2k+0 & -4+k(k-1)+0 \\ 1+2k-1 & 2+k(k-1)+2k \end{pmatrix} = \begin{pmatrix} 2k-2 & k^2-k-4 \\ 2k & k^2+k+2 \end{pmatrix} \end{aligned}$$

As $\mathbf{AB}$ is symmetric, we must have $2k = k^2-k-4 \Rightarrow 0 = k^2-3k-4 \Rightarrow 0 = (k-4)(k+1) \Rightarrow k = -1$ or $k = 4$.

When $k = -1$, $\mathbf{AB} = \begin{pmatrix} -4 & -2 \\ -2 & 2 \end{pmatrix}$.

When $k = 4$, $\mathbf{AB} = \begin{pmatrix} 6 & 8 \\ 8 & 22 \end{pmatrix}$.

34 a)

$$\mathbf{AB} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{pmatrix} \begin{pmatrix} 2 & 3 & 1 \\ 1 & 0 & 4 \\ 0 & 4 & 5 \end{pmatrix} = \begin{pmatrix} 2 & 11 & 11 \\ 3 & 0 & 12 \\ 8 & 32 & 29 \end{pmatrix}$$

34 b)

$$\mathbf{BA} = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 0 & 4 \\ 0 & 4 & 5 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 6 & 9 & 9 \\ 17 & 0 & 22 \\ 20 & 12 & 25 \end{pmatrix}$$

35 a)

$$\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}^3 = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$$

$$= \begin{pmatrix} a^2 & 0 & 0 \\ 0 & b^2 & 0 \\ 0 & 0 & c^2 \end{pmatrix} \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} = \begin{pmatrix} a^3 & 0 & 0 \\ 0 & b^3 & 0 \\ 0 & 0 & c^3 \end{pmatrix}$$

35 b)

$$\mathbf{A}^3 = \begin{pmatrix} \sin\theta & 0 & 0 \\ 0 & -\cos\theta & 0 \\ 0 & 0 & -\sin\theta \end{pmatrix}^3 = \begin{pmatrix} (\sin\theta)^3 & 0 & 0 \\ 0 & (-\cos\theta)^3 & 0 \\ 0 & 0 & (-\sin\theta)^3 \end{pmatrix}$$

$$= \begin{pmatrix} \sin^3\theta & 0 & 0 \\ 0 & -\cos^3\theta & 0 \\ 0 & 0 & -\sin^3\theta \end{pmatrix}$$

36)

Let $\mathbf{A} = \begin{pmatrix} a \\ b \end{pmatrix}$.

$$\begin{pmatrix} 2 & 3 \\ -3 & 4 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 12 \\ -69 \end{pmatrix} \Rightarrow \begin{pmatrix} 2a+3b \\ -3a+4b \end{pmatrix} = \begin{pmatrix} 12 \\ -69 \end{pmatrix}$$

$$2a + 3b = 12 \quad (1) \qquad -3a + 4b = -69 \quad (2)$$

$$\therefore a = 15 \text{ and } b = -6 \qquad \therefore \mathbf{A} = \begin{pmatrix} 15 \\ -6 \end{pmatrix}$$

37)

Let $\mathbf{A} = \begin{pmatrix} a & b & c \end{pmatrix}$.

$$\begin{pmatrix} a & b & c \end{pmatrix} \begin{pmatrix} -4 & 1 \\ 0 & 0 \\ 4 & 5 \end{pmatrix} = \begin{pmatrix} 32 & 24 \end{pmatrix} \Rightarrow \begin{pmatrix} -4a+4c & a+5c \end{pmatrix} = \begin{pmatrix} 32 & 24 \end{pmatrix}$$

$$-4a + 4c = 32 \quad (1) \qquad a + 5c = 24 \quad (2)$$

$$\therefore a = -\frac{8}{3} \text{ and } c = \frac{16}{3} \qquad \therefore \mathbf{A} = \begin{pmatrix} -\frac{8}{3} & b & \frac{16}{3} \end{pmatrix} \text{ where } b \in \mathbb{R}$$

So a possible matrix could be $\mathbf{A} = \begin{pmatrix} -\frac{8}{3} & 1 & \frac{16}{3} \end{pmatrix}$.

38)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

$$\begin{pmatrix} 2 & -3 \\ 0 & 6 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 13 & 1 \\ -6 & -18 \end{pmatrix} \Rightarrow \begin{pmatrix} 2a-3c & 2b-3d \\ 6c & 6d \end{pmatrix} = \begin{pmatrix} 13 & 1 \\ -6 & -18 \end{pmatrix}$$

$$6c = -6 \Rightarrow c = -1 \qquad 6d = -18 \Rightarrow d = -3$$

$$2a - 3(-1) = 13 \Rightarrow a = 5 \qquad 2b - 3(-3) = 1 \Rightarrow b = -4$$

$$\therefore \mathbf{A} = \begin{pmatrix} 5 & -4 \\ -1 & -3 \end{pmatrix}$$

39)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} -5 & 2 & -4 \\ 10 & 2 & 16 \end{pmatrix} = \begin{pmatrix} -15 & 6 & -12 \\ 15 & 6 & 28 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -5a+10b & 2a+2b & -4a+16b \\ -5c+10d & 2c+2d & -4c+16d \end{pmatrix} = \begin{pmatrix} -15 & 6 & -12 \\ 15 & 6 & 28 \end{pmatrix}$$

$$-5a + 10b = -15\ (1) \quad 2a + 2b = 6\ (2) \quad -4a + 16b = -12\ (3)$$

$$\therefore a = 3 \text{ and } b = 0$$

$$-5c + 10d = 15\ (4) \quad 2c + 2d = 6\ (5) \quad -4c + 16d = 28\ (6)$$

$$\therefore c = 1 \text{ and } d = 2 \qquad \therefore \mathbf{A} = \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}$$

40)

$$\mathbf{AB} = \begin{pmatrix} 6 & 2 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 6\times2+2\times0 & 6\times0+2\times2 \\ 3\times2+1\times0 & 3\times0+1\times2 \end{pmatrix} = \begin{pmatrix} 12 & 4 \\ 6 & 2 \end{pmatrix}$$

$$\mathbf{BA} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 6 & 2 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 2\times6+0\times3 & 2\times2+0\times1 \\ 0\times6+2\times3 & 0\times2+2\times1 \end{pmatrix} = \begin{pmatrix} 12 & 4 \\ 6 & 2 \end{pmatrix}$$

As $\mathbf{AB} = \mathbf{BA}$, $\mathbf{A}$ and $\mathbf{B}$ are commutative under multiplication.

41)

$$\mathbf{AB} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix} = \begin{pmatrix} 1\times3+1\times3 & 1\times0+1\times3 \\ 0\times3+1\times3 & 0\times0+1\times3 \end{pmatrix} = \begin{pmatrix} 6 & 3 \\ 3 & 3 \end{pmatrix}$$

$$\mathbf{BA} = \begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3\times1+0\times0 & 3\times1+0\times1 \\ 3\times1+3\times0 & 3\times1+3\times1 \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ 3 & 6 \end{pmatrix}$$

As $\mathbf{AB} \neq \mathbf{BA}$, matrix multiplication is not, in general, commutative.

42)

$$\mathbf{A}(\mathbf{BC}) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \left(\begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix} \begin{pmatrix} 5 & 5 \\ 5 & 0 \end{pmatrix} \right) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 15 & 15 \\ 30 & 15 \end{pmatrix} = \begin{pmatrix} 45 & 30 \\ 30 & 15 \end{pmatrix}$$

$$(\mathbf{AB})\mathbf{C} = \left(\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix} \right) \begin{pmatrix} 5 & 5 \\ 5 & 0 \end{pmatrix} = \begin{pmatrix} 6 & 3 \\ 3 & 3 \end{pmatrix} \begin{pmatrix} 5 & 5 \\ 5 & 0 \end{pmatrix} = \begin{pmatrix} 45 & 30 \\ 30 & 15 \end{pmatrix}$$

As $\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$, $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are associative under multiplication.

43 a)

$$(\mathbf{AB})\mathbf{C} = \left(\begin{pmatrix} k & 3 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} -2 & 0 \\ 1 & 4 \end{pmatrix} \right) \begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} -2k+3 & 12 \\ -5 & -4 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} -2k+3 & 4k+30 \\ -5 & -2 \end{pmatrix}$$

$$\mathbf{A}(\mathbf{BC}) = \begin{pmatrix} k & 3 \\ 2 & -1 \end{pmatrix} \left(\begin{pmatrix} -2 & 0 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix} \right) = \begin{pmatrix} k & 3 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} -2 & 4 \\ 1 & 10 \end{pmatrix} = \begin{pmatrix} -2k+3 & 4k+30 \\ -5 & -2 \end{pmatrix}$$

$$\therefore (\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$$

43 b)

This shows that $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are associative under matrix multiplication.

44)

$$(\mathbf{AB})^2 = \left(\begin{pmatrix} 5 & 2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} -2 & 1 \\ 0 & 4 \end{pmatrix} \right)^2 = \begin{pmatrix} -10 & 13 \\ -2 & 1 \end{pmatrix}^2 = \begin{pmatrix} 74 & -117 \\ 18 & -25 \end{pmatrix}$$

$$\mathbf{A}^2\mathbf{B}^2 = \begin{pmatrix} 5 & 2 \\ 1 & 0 \end{pmatrix}^2 \begin{pmatrix} -2 & 1 \\ 0 & 4 \end{pmatrix}^2 = \begin{pmatrix} 27 & 10 \\ 5 & 2 \end{pmatrix} \begin{pmatrix} 4 & 2 \\ 0 & 16 \end{pmatrix} = \begin{pmatrix} 108 & 214 \\ 20 & 42 \end{pmatrix}$$

$$\therefore (\mathbf{AB})^2 \neq \mathbf{A}^2\mathbf{B}^2$$

45)

$$\begin{pmatrix} 5a & -3 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} p \\ q \end{pmatrix} = 4 \begin{pmatrix} p \\ q \end{pmatrix} \Rightarrow \begin{pmatrix} 5ap - 3q \\ p + 2q \end{pmatrix} = \begin{pmatrix} 4p \\ 4q \end{pmatrix}$$

$$5ap - 3q = 4p \quad (1) \qquad p + 2q = 4q \quad (2)$$

$$(2) \Rightarrow p = 2q$$

$$(1) \Rightarrow 5a \times 2q - 3q = 4 \times 2q \Rightarrow 10a - 3 = 8 \Rightarrow a = \frac{11}{10}$$

46)

$$\begin{pmatrix} a & 1 & 2 \\ 0 & 3 & a \\ a & 2 & 1 \end{pmatrix}^2 = \begin{pmatrix} 15 & 2 & -13 \\ 25 & -1 & -20 \\ 20 & 3 & -19 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a^2+2a & a+7 & 3a+2 \\ a^2 & 9+2a & 4a \\ a^2+a & a+8 & 4a+1 \end{pmatrix} = \begin{pmatrix} 15 & 2 & -13 \\ 25 & -1 & -20 \\ 20 & 3 & -19 \end{pmatrix}$$

$$a^2+2a = 15 \Rightarrow a = -5 \text{ or } a = 3$$

$$a^2 = 25 \Rightarrow a = 5 \text{ or } a = -5$$

$$a^2+a = 20 \Rightarrow a = -5 \text{ or } a = 4$$

$$a+7 = 2 \Rightarrow a = -5$$

$$9+2a = -1 \Rightarrow a = -5$$

$$a+8 = 3 \Rightarrow a = -5$$

$$3a+2 = -13 \Rightarrow a = -5$$

$$4a = -20 \Rightarrow a = -5$$

$$4a+1 = -19 \Rightarrow a = -5$$

$$\therefore a = -5$$

47)

$$\begin{pmatrix} 2 & 2 & -1 \\ 2 & 1 & -2 \\ -2 & -2 & 1 \end{pmatrix} \begin{pmatrix} 9 & 6 & 3 \\ 2 & 0 & x \\ -2 & -6 & 4 \end{pmatrix} = \begin{pmatrix} 9 & 6 & 3 \\ 2 & 0 & x \\ -2 & -6 & 4 \end{pmatrix} \begin{pmatrix} 2 & 2 & -1 \\ 2 & 1 & -2 \\ -2 & -2 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 24 & 18 & 2x+2 \\ 24 & 24 & x-2 \\ -24 & -18 & -2x-2 \end{pmatrix} = \begin{pmatrix} 24 & 18 & -18 \\ 4-2x & 4-2x & x-2 \\ -24 & -18 & 18 \end{pmatrix}$$

$$2x+2 = -18 \Rightarrow x = -10$$

$$4-2x = 24 \Rightarrow x = -10$$

$$\therefore x = -10$$

48 a)

$$\mathbf{A}^2 = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 8 & 1 \end{pmatrix}$$

48 b)

$$\mathbf{A}^3 = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}^3 = \begin{pmatrix} 1 & 0 \\ 12 & 1 \end{pmatrix}$$

48 c)

$$\mathbf{A}^4 = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}^4 = \begin{pmatrix} 1 & 0 \\ 16 & 1 \end{pmatrix}$$

48 d)

$$\mathbf{A}^k = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}^k = \begin{pmatrix} 1 & 0 \\ 4k & 1 \end{pmatrix}$$

49 a)

$$\mathbf{A}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}^2 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$$

49 b)

$$\mathbf{A}^3 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}^3 = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$$

49 c)

$$\mathbf{A}^4 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}^4 = \begin{pmatrix} 8 & 8 \\ 8 & 8 \end{pmatrix}$$

49 d)

$$\mathbf{A}^k = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}^k = \begin{pmatrix} 2^{k-1} & 2^{k-1} \\ 2^{k-1} & 2^{k-1} \end{pmatrix} = 2^{k-1} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

50 a)

$$\mathbf{A}^2 = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}^2 = \begin{pmatrix} 3 & 2 \\ -2 & -1 \end{pmatrix}$$

50 b)

$$\mathbf{A}^3 = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}^3 = \begin{pmatrix} 4 & 3 \\ -3 & -2 \end{pmatrix}$$

50 c)

$$\mathbf{A}^4 = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}^4 = \begin{pmatrix} 5 & 4 \\ -4 & -3 \end{pmatrix}$$

50 d)

$$\mathbf{A}^k = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}^k = \begin{pmatrix} k+1 & k \\ -k & 1-k \end{pmatrix}$$

51)

$$\begin{pmatrix} 2 & a \\ 2b & 1 \end{pmatrix}\begin{pmatrix} 3a & 4 \\ 2 & 2b \end{pmatrix} = \begin{pmatrix} 48 & 116 \\ 326 & 90 \end{pmatrix} \Rightarrow \begin{pmatrix} 8a & 8+2ab \\ 6ab+2 & 10b \end{pmatrix} = \begin{pmatrix} 48 & 116 \\ 326 & 90 \end{pmatrix}$$

$$8a = 48 \Rightarrow a = 6 \qquad 10b = 90 \Rightarrow b = 9$$

Check: $8 + 2ab = 8 + 2 \times 6 \times 9 = 116$.

Check: $6ab + 2 = 6 \times 6 \times 9 + 2 = 326$.

$$\therefore a = 6 \text{ and } b = 9$$

52)

$$\begin{pmatrix} 3a & -2 \\ 5a & -3 \end{pmatrix}\begin{pmatrix} 8 & -3a \\ 4a & 1 \end{pmatrix} = \begin{pmatrix} b+2 & 22-2b \\ 2b-4 & -2b-3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 16a & -9a^2-2 \\ 28a & -15a^2-3 \end{pmatrix} = \begin{pmatrix} b+2 & 22-2b \\ 2b-4 & -2b-3 \end{pmatrix}$$

$$16a = b + 2 \Rightarrow 16a - b = 2 \quad (1) \qquad 28a = 2b - 4 \Rightarrow 28a - 2b = -4 \quad (2)$$

$$\therefore a = 2 \text{ and } b = 30$$

$$-9a^2 - 2 = 22 - 2b \Rightarrow -9a^2 + 2b = 24$$

Check: $-9a^2 + 2b = -9 \times 2^2 + 2 \times 30 = 24$.

$$-15a^2 - 3 = -2b - 3 \Rightarrow -15a^2 + 2b = 0$$

Check: $-15a^2 + 2b = -15 \times 2^2 + 2 \times 30 = 0$.

$$\therefore a = 2 \text{ and } b = 30$$

53)

$$\begin{pmatrix} b & 5 \\ 2a & 3 \end{pmatrix}\begin{pmatrix} b & 5 \\ 2a & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} b^2+10a & 5b+15 \\ 2ab+6a & 10a+9 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$10a + 9 = 1 \Rightarrow a = -\frac{4}{5} \qquad 5b + 15 = 0 \Rightarrow b = -3$$

Check: $b^2 + 10a = (-3)^2 + 10 \times -\frac{4}{5} = 1$.

Check: $2ab + 6a = 2 \times -\frac{4}{5} \times -3 + 6 \times -\frac{4}{5} = 0$.

$$\therefore a = -\frac{4}{5} \text{ and } b = -3$$

54)

$$\begin{pmatrix} 3b & 4 \\ 5a & 2 \end{pmatrix}^2 = \mathbf{0} \Rightarrow \begin{pmatrix} 20a+9b^2 & 12b+8 \\ 15ab+10a & 20a+4 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$20a + 4 = 0 \Rightarrow a = -\frac{1}{5} \qquad 12b + 8 = 0 \Rightarrow b = -\frac{2}{3}$$

55)

$$p\begin{pmatrix}2 & 5\\ 10 & 2\end{pmatrix}-\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}=q\begin{pmatrix}7 & 2\\ 4 & 7\end{pmatrix}\Rightarrow\begin{pmatrix}2p-1 & 5p\\ 10p & 2p-1\end{pmatrix}=\begin{pmatrix}7q & 2q\\ 4q & 7q\end{pmatrix}$$

$$2p-1=7q\Rightarrow 2p-7q=1\quad(1)\qquad 5p=2q\Rightarrow 5p-2q=0\quad(2)$$

$$\therefore p=-\frac{2}{31}\text{ and }q=-\frac{5}{31}$$

56)

$$p\begin{pmatrix}1 & 10\\ 50 & 1\end{pmatrix}+q\begin{pmatrix}1 & 3\\ 15 & 1\end{pmatrix}=5\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}p & 10p\\ 50p & p\end{pmatrix}+\begin{pmatrix}q & 3q\\ 15q & q\end{pmatrix}=\begin{pmatrix}5 & 0\\ 0 & 5\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}p+q & 10p+3q\\ 50p+15q & p+q\end{pmatrix}=\begin{pmatrix}5 & 0\\ 0 & 5\end{pmatrix}$$

$$p+q=5\quad(1)\qquad 10p+3q=0\quad(2)$$

$$\therefore p=-\frac{15}{7}\text{ and }q=\frac{50}{7}$$

57)

$$\begin{pmatrix}5 & 1\\ x & 2\\ 0 & 1\end{pmatrix}\begin{pmatrix}2y & 3z\\ z & y\end{pmatrix}=\begin{pmatrix}38 & 123\\ 52 & 150\\ 8 & 3\end{pmatrix}$$

$$=\begin{pmatrix}10y+z & 15z+y\\ 2xy+2z & 3xz+2y\\ z & y\end{pmatrix}=\begin{pmatrix}38 & 123\\ 52 & 150\\ 8 & 3\end{pmatrix}$$

$$\therefore z=8\text{ and }y=3$$

$$2xy+2z=52\Rightarrow 2x\times 3+2\times 8=52\Rightarrow 6x+16=52\Rightarrow x=6$$

Check: $10y+z=10\times 3+8=38$.

Check: $15z+y=15\times 8+3=123$.

Check: $3xz+2y=3\times 6\times 8+2\times 3=150$.

$$\therefore x=6,\ y=3\text{ and }z=8$$

58)

$$\begin{pmatrix}-1 & x & 3 & 4\\ x & 3 & y & y\end{pmatrix}\begin{pmatrix}4 & 5\\ 2 & 3\\ x & y\\ 4 & 2\end{pmatrix}=2\mathbf{I}$$

$$\Rightarrow\begin{pmatrix}5x+12 & 3x+3y+3\\ 4x+6+xy+4y & 5x+9+y^2+2y\end{pmatrix}=\begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}$$

$$5x + 12 = 2 \Rightarrow x = -2$$

$$3 \times (-2) + 3y + 3 = 0 \Rightarrow y = 1$$

Check: $4x + 6 + xy + 4y = 4 \times (-2) + 6 - 2 \times 1 + 4 \times 1 = 0$.

Check: $5x + 9 + y^2 + 2y = 5 \times (-2) + 9 + 1^2 + 2 \times 1 = 2$.

$$\therefore x = -2 \text{ and } y = 1$$

59)

$$\begin{pmatrix} 5 & 3 \\ 18 & 2 \end{pmatrix} \begin{pmatrix} x+3 & 1 \\ x-2 & 10 \end{pmatrix} = \begin{pmatrix} x+3 & 1 \\ x-2 & 10 \end{pmatrix} \begin{pmatrix} 5 & 3 \\ 18 & 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 5(x+3)+3(x-2) & 35 \\ 18(x+3)+2(x-2) & 38 \end{pmatrix} = \begin{pmatrix} 5(x+3)+18 & 3(x+3)+2 \\ 5(x-2)+180 & 3(x-2)+20 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 8x+9 & 35 \\ 20x+50 & 38 \end{pmatrix} = \begin{pmatrix} 5x+33 & 3x+11 \\ 5x+170 & 3x+14 \end{pmatrix}$$

$$8x + 9 = 5x + 33 \Rightarrow 3x = 24 \Rightarrow x = 8$$

$$35 = 3x + 11 \Rightarrow 3x = 24 \Rightarrow x = 8$$

$$20x + 50 = 5x + 170 \Rightarrow 15x = 120 \Rightarrow x = 8$$

$$38 = 3x + 14 \Rightarrow 3x = 24 \Rightarrow x = 8$$

$$\therefore x = 8$$

60)

$$\begin{pmatrix} k & 0 \\ k+2 & 3-k \end{pmatrix} \begin{pmatrix} 2 & 0 \\ -k & 3 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ -k & 3 \end{pmatrix} \begin{pmatrix} k & 0 \\ k+2 & 3-k \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2k & 0 \\ k^2-k+4 & 9-3k \end{pmatrix} = \begin{pmatrix} 2k & 0 \\ -k^2+3k+6 & 9-3k \end{pmatrix}$$

$$k^2 - k + 4 = -k^2 + 3k + 6 \Rightarrow 2k^2 - 4k - 2 = 0 \Rightarrow k^2 - 2k - 1 = 0$$

$$\Rightarrow (k-1)^2 - 1 - 1 = 0 \Rightarrow (k-1)^2 = 2 \Rightarrow k = 1 \pm \sqrt{2}$$

61)

$$\mathbf{AB} = \begin{pmatrix} 2x^2 & 16x \\ 8x & x \end{pmatrix} \begin{pmatrix} x+1 & 16 \\ 8 & 2-x \end{pmatrix} = \begin{pmatrix} 2x^2(x+1)+128x & 32x^2+16x(2-x) \\ 8x(x+1)+8x & 128x+x(2-x) \end{pmatrix}$$

$$= \begin{pmatrix} 2x^3+2x^2+128x & 16x^2+32x \\ 8x^2+16x & 130x-x^2 \end{pmatrix}$$

$$\mathbf{BA} = \begin{pmatrix} x+1 & 16 \\ 8 & 2-x \end{pmatrix} \begin{pmatrix} 2x^2 & 16x \\ 8x & x \end{pmatrix} = \begin{pmatrix} 2x^2(x+1)+128x & 16x(x+1)+16x \\ 16x^2+8x(2-x) & 128x+x(2-x) \end{pmatrix}$$

$$= \begin{pmatrix} 2x^3+2x^2+128x & 16x^2+32x \\ 8x^2+16x & 130x-x^2 \end{pmatrix}$$

$\mathbf{AB} = \mathbf{BA}$ so the matrices $\mathbf{A}$ and $\mathbf{B}$ are commutative under multiplication.

62)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

$$4\begin{pmatrix} a & b \\ c & d \end{pmatrix} - 2\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix} \Rightarrow \begin{pmatrix} 4a-2 & 4b \\ 4c & 4d-2 \end{pmatrix} = \begin{pmatrix} a-3b & 2a+4b \\ c-3d & 2c+4d \end{pmatrix}$$

$$4a - 2 = a - 3b \Rightarrow 3a + 3b = 2 \qquad 4b = 2a + 4b \Rightarrow 2a = 0 \Rightarrow a = 0$$

$$\therefore 3 \times 0 + 3b = 2 \Rightarrow b = \frac{2}{3}$$

$$4c = c - 3d \Rightarrow 3c + 3d = 0 \Rightarrow c + d = 0$$

$$4d - 2 = 2c + 4d \Rightarrow 2c = -2 \Rightarrow c = -1$$

$$\therefore -1 + d = 0 \Rightarrow d = 1 \qquad \therefore \mathbf{A} = \begin{pmatrix} 0 & \frac{2}{3} \\ -1 & 1 \end{pmatrix}$$

63)

$$k\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a+3 & -2a \\ 6-a & a \end{pmatrix}\begin{pmatrix} a-6 & a-3 \\ 10-a & a-5 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} = \begin{pmatrix} (a+3)(a-6) - 2a(10-a) & (a+3)(a-3) - 2a(a-5) \\ (6-a)(a-6) + a(10-a) & (6-a)(a-3) + a(a-5) \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} = \begin{pmatrix} 3a^2 - 23a - 18 & -a^2 + 10a - 9 \\ -2a^2 + 22a - 36 & 4a - 18 \end{pmatrix}$$

$$-a^2 + 10a - 9 = 0 \Rightarrow a = 9 \text{ or } a = 1$$

$$-2a^2 + 22a - 36 = 0 \Rightarrow a = 9 \text{ or } a = 2$$

$$\therefore a = 9 \qquad k = 3 \times 9^2 - 23 \times 9 - 18 = 18$$

Check: $4 \times 9 - 18 = 18$.

$$\therefore a = 9 \text{ and } k = 18$$

64)

$$\begin{pmatrix} 1 & -1 & -3 \\ 0 & x & 1 \\ 3 & -2 & 1 \end{pmatrix}\begin{pmatrix} \frac{4}{3} & 1 & -\frac{5}{3} \\ 2 & x+13 & 3 \\ 1 & -8 & 0 \end{pmatrix} = \begin{pmatrix} \frac{4}{3} & 1 & -\frac{5}{3} \\ 2 & x+13 & 3 \\ 1 & -8 & 0 \end{pmatrix}\begin{pmatrix} 1 & -1 & -3 \\ 0 & x & 1 \\ 3 & -2 & 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -\frac{11}{3} & 12 - x & -\frac{14}{3} \\ 2x+1 & x^2 + 13x - 8 & 3x \\ 1 & -2x - 31 & -11 \end{pmatrix} = \begin{pmatrix} -\frac{11}{3} & x+2 & -\frac{14}{3} \\ 11 & x^2 + 13x - 8 & x+10 \\ 1 & -8x - 1 & -11 \end{pmatrix}$$

$$12 - x = x + 2 \Rightarrow 2x = 10 \Rightarrow x = 5$$

$$2x + 1 = 11 \Rightarrow 2x = 10 \Rightarrow x = 5$$

$$3x = x + 10 \Rightarrow 2x = 10 \Rightarrow x = 5$$

$$-2x - 31 = -8x - 1 \Rightarrow 6x = 30 \Rightarrow x = 5$$

$$\therefore x = 5$$

65)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$.

$$\mathbf{A} + \mathbf{B} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}$$

$$\mathbf{B} + \mathbf{A} = \begin{pmatrix} e & f \\ g & h \end{pmatrix} + \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} e+a & f+b \\ g+c & h+d \end{pmatrix} = \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}$$

So $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$.

$\therefore$ matrix addition is commutative for 2×2 matrices

66)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$.

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} - \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} a-e & b-f \\ c-g & d-h \end{pmatrix}$$

$$\mathbf{B} - \mathbf{A} = \begin{pmatrix} e & f \\ g & h \end{pmatrix} - \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} e-a & f-b \\ g-c & h-d \end{pmatrix}$$

In general, $a - e \neq e - a$, $b - f \neq f - b$, $c - g \neq g - c$, $d - h \neq h - d$.

So $\mathbf{A} - \mathbf{B} \neq \mathbf{B} - \mathbf{A}$.

$\therefore$ matrix subtraction is not commutative for 2×2 matrices

67)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$.

$$\mathbf{AB} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}$$

$$\mathbf{BA} = \begin{pmatrix} e & f \\ g & h \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} ae+cf & be+df \\ ag+ch & bg+dh \end{pmatrix}$$

In general, $ae + bg \neq ae + cf$, $af + bh \neq be + df$, $ce + dg \neq ag + ch$, $cf + dh \neq bg + dh$.

So $\mathbf{AB} \neq \mathbf{BA}$.

$\therefore$ matrix multiplication is not in general commutative for 2×2 matrices

68 a)

$\mathbf{A}^T = \begin{pmatrix} 2 & 6 \\ 3 & -7 \end{pmatrix}$

68 b)

$$\mathbf{B}^T = \begin{pmatrix} 4 & -9 \\ 2 & 1 \\ 0 & 4 \end{pmatrix}$$

68 c)

$$\mathbf{C}^T = \begin{pmatrix} 1 & 10 & 3 \\ 2 & -9 & 0 \\ -7 & 2 & 8 \end{pmatrix}$$

69)

$$\mathbf{A} = \begin{pmatrix} 5 & -8 & 5 \\ 2 & 1 & 3 \end{pmatrix}, \quad \mathbf{A}^T = \begin{pmatrix} 5 & 2 \\ -8 & 1 \\ 5 & 3 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 8 & -5 \\ 2 & -1 \end{pmatrix}, \quad \mathbf{B}^T = \begin{pmatrix} 8 & 2 \\ -5 & -1 \end{pmatrix}$$

$$\mathbf{AA}^T,\ \mathbf{A}^T\mathbf{A},\ \mathbf{A}^T\mathbf{B},\ \mathbf{A}^T\mathbf{B}^T,\ \mathbf{BA},\ \mathbf{BB}^T,\ \mathbf{B}^T\mathbf{A},\ \mathbf{B}^T\mathbf{B}$$

70 a)

$$\mathbf{AA}^T = \begin{pmatrix} k & 2k \\ 4 & 6k \end{pmatrix} \begin{pmatrix} k & 2k \\ 4 & 6k \end{pmatrix}^T = \begin{pmatrix} k & 2k \\ 4 & 6k \end{pmatrix} \begin{pmatrix} k & 4 \\ 2k & 6k \end{pmatrix} = \begin{pmatrix} 45 & 96 \\ 96 & 340 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 5k^2 & 12k^2 + 4k \\ 12k^2 + 4k & 36k^2 + 16 \end{pmatrix} = \begin{pmatrix} 45 & 96 \\ 96 & 340 \end{pmatrix}$$

$$5k^2 = 45 \Rightarrow k^2 = 9 \Rightarrow k = \pm 3$$

$$12k^2 + 4k = 96 \Rightarrow 12k^2 + 4k - 96 = 0 \Rightarrow k = -3 \text{ or } k = \frac{8}{3}$$

$$\therefore k = -3$$

70 b)

$$\mathbf{A}^T\mathbf{A} = \begin{pmatrix} k & 2k \\ 4 & 6k \end{pmatrix}^T \begin{pmatrix} k & 2k \\ 4 & 6k \end{pmatrix} = \begin{pmatrix} k & 4 \\ 2k & 6k \end{pmatrix} \begin{pmatrix} k & 2k \\ 4 & 6k \end{pmatrix} = 32 \begin{pmatrix} 1 & 4 \\ 4 & 20 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} k^2 + 16 & 2k^2 + 24k \\ 2k^2 + 24k & 40k^2 \end{pmatrix} = \begin{pmatrix} 32 & 128 \\ 128 & 640 \end{pmatrix}$$

$$k^2 + 16 = 32 \Rightarrow k^2 = 16 \Rightarrow k = \pm 4$$

$$2k^2 + 24k = 128 \Rightarrow 2k^2 + 24k - 128 = 0 \Rightarrow k = -16 \text{ or } k = 4$$

$$\therefore k = 4$$

71 a)

$$\mathbf{A}\mathbf{A}^T\mathbf{A} = \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^T \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & k & 0 \\ k & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 3k^2+2 & k^3+4k & k^2+3 \\ k^3+3k & 3k^2+1 & 2k \\ 1 & k & 2 \end{pmatrix}$$

71 b)

$$\mathbf{A}^T\mathbf{A}\mathbf{A}^T = \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^T \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^T$$

$$= \begin{pmatrix} 1 & k & 0 \\ k & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & k & 0 \\ k & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 3k^2+2 & k^3+3k & 1 \\ k^3+4k & 3k^2+1 & k \\ k^2+3 & 2k & 2 \end{pmatrix}$$

72)

$$(\mathbf{AB})^T = \left(\begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 5 & 3 \\ 1 & 2 \end{pmatrix} \right)^T = \begin{pmatrix} 16 & 11 \\ 2 & 4 \end{pmatrix}^T = \begin{pmatrix} 16 & 2 \\ 11 & 4 \end{pmatrix}$$

$$\mathbf{B}^T\mathbf{A}^T = \begin{pmatrix} 5 & 3 \\ 1 & 2 \end{pmatrix}^T \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}^T = \begin{pmatrix} 5 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 16 & 2 \\ 11 & 4 \end{pmatrix}$$

$$\therefore (\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T$$

73)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

$$(\mathbf{AB})^T = \mathbf{B}^T\mathbf{A}^T \Rightarrow \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix} \right)^T = \begin{pmatrix} 12 & 5 \\ 7 & 3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2a+3b & a+b \\ 2c+3d & c+d \end{pmatrix}^T = \begin{pmatrix} 12 & 5 \\ 7 & 3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2a+3b & 2c+3d \\ a+b & c+d \end{pmatrix} = \begin{pmatrix} 12 & 5 \\ 7 & 3 \end{pmatrix}$$

$$2a+3b = 12 \quad (1) \qquad a+b = 7 \quad (2)$$

Solve (1) and (2) to get $a = 9$ and $b = -2$.

$$2c+3d = 5 \quad (3) \qquad c+d = 3 \quad (4)$$

Solve (3) and (4) to get $c = 4$ and $d = -1$.

$$\therefore \mathbf{A} = \begin{pmatrix} 9 & -2 \\ 4 & -1 \end{pmatrix}$$

74)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$.

$$(\mathbf{AB})^T = \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} \right)^T = \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}^T = \begin{pmatrix} ae+bg & ce+dg \\ af+bh & cf+dh \end{pmatrix}$$

$$\mathbf{B}^T\mathbf{A}^T = \begin{pmatrix} e & f \\ g & h \end{pmatrix}^T \begin{pmatrix} a & b \\ c & d \end{pmatrix}^T = \begin{pmatrix} e & g \\ f & h \end{pmatrix} \begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} ae+bg & ce+dg \\ af+bh & cf+dh \end{pmatrix}$$

$$\therefore (\mathbf{AB})^T \equiv \mathbf{B}^T\mathbf{A}^T \text{ for } 2 \times 2 \text{ matrices}$$

75)

$$(\mathbf{AB}^T - \mathbf{BA}^T)^2 = -25\mathbf{I}$$

$$\Rightarrow \left(\begin{pmatrix} x & 2 \\ 1 & y \end{pmatrix} \begin{pmatrix} 3 & x \\ 2y & 1 \end{pmatrix}^T - \begin{pmatrix} 3 & x \\ 2y & 1 \end{pmatrix} \begin{pmatrix} x & 2 \\ 1 & y \end{pmatrix}^T \right)^2 = \begin{pmatrix} -25 & 0 \\ 0 & -25 \end{pmatrix}$$

$$\Rightarrow \left(\begin{pmatrix} x & 2 \\ 1 & y \end{pmatrix} \begin{pmatrix} 3 & 2y \\ x & 1 \end{pmatrix} - \begin{pmatrix} 3 & x \\ 2y & 1 \end{pmatrix} \begin{pmatrix} x & 1 \\ 2 & y \end{pmatrix} \right)^2 = \begin{pmatrix} -25 & 0 \\ 0 & -25 \end{pmatrix}$$

$$\Rightarrow \left(\begin{pmatrix} 5x & 2xy+2 \\ xy+3 & 3y \end{pmatrix} - \begin{pmatrix} 5x & xy+3 \\ 2xy+2 & 3y \end{pmatrix} \right)^2 = \begin{pmatrix} -25 & 0 \\ 0 & -25 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & xy-1 \\ 1-xy & 0 \end{pmatrix}^2 = \begin{pmatrix} -25 & 0 \\ 0 & -25 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} (xy-1)(1-xy) & 0 \\ 0 & (xy-1)(1-xy) \end{pmatrix} = \begin{pmatrix} -25 & 0 \\ 0 & -25 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -(xy-1)(xy-1) & 0 \\ 0 & -(xy-1)(xy-1) \end{pmatrix} = \begin{pmatrix} -25 & 0 \\ 0 & -25 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -(xy-1)^2 & 0 \\ 0 & -(xy-1)^2 \end{pmatrix} = \begin{pmatrix} -25 & 0 \\ 0 & -25 \end{pmatrix}$$

$$-(xy-1)^2 = -25 \Rightarrow (xy-1)^2 = 25 \Rightarrow xy - 1 = \pm 5 \Rightarrow xy = 1 \pm 5$$

x and y are both positive, so we must have that $xy = 6$.

x and y are both integers greater than 1, and $x < y$, so $x = 2$ and $y = 3$.

76)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$.

$$(\mathbf{A}+\mathbf{B})^T = \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} e & f \\ g & h \end{pmatrix} \right)^T = \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}^T = \begin{pmatrix} a+e & c+g \\ b+f & d+h \end{pmatrix}$$

$$\mathbf{A}^T + \mathbf{B}^T = \begin{pmatrix} a & b \\ c & d \end{pmatrix}^T + \begin{pmatrix} e & f \\ g & h \end{pmatrix}^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix} + \begin{pmatrix} e & g \\ f & h \end{pmatrix} = \begin{pmatrix} a+e & c+g \\ b+f & d+h \end{pmatrix}$$

$$\therefore (\mathbf{A}+\mathbf{B})^T \equiv \mathbf{A}^T + \mathbf{B}^T \text{ for } 2 \times 2 \text{ matrices}$$

77)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

$$(k\mathbf{A})^T = \left(k \begin{pmatrix} a & b \\ c & d \end{pmatrix} \right)^T = \begin{pmatrix} ka & kb \\ kc & kd \end{pmatrix}^T = \begin{pmatrix} ka & kc \\ kb & kd \end{pmatrix}$$

$$k\mathbf{A}^T = k \begin{pmatrix} a & b \\ c & d \end{pmatrix}^T = k \begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} ka & kc \\ kb & kd \end{pmatrix}$$

$$\therefore (k\mathbf{A})^T \equiv k\mathbf{A}^T \text{ for } 2 \times 2 \text{ matrices}$$

78)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

$$\det \mathbf{A} = a \times d - b \times c = ad - bc$$

$$\mathbf{A}^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$$

$$\det(\mathbf{A}^T) = a \times d - c \times b = ad - bc$$

$$\therefore \det(\mathbf{A}^T) \equiv \det \mathbf{A} \text{ for } 2 \times 2 \text{ matrices}$$

79)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

$$\mathbf{A}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} \frac{d}{ad-bc} & -\frac{b}{ad-bc} \\ -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix}$$

$$(\mathbf{A}^{-1})^T = \begin{pmatrix} \frac{d}{ad-bc} & -\frac{c}{ad-bc} \\ -\frac{b}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix}$$

$$\mathbf{A}^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$$

$$(\mathbf{A}^T)^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -c \\ -b & a \end{pmatrix} = \begin{pmatrix} \frac{d}{ad-bc} & -\frac{c}{ad-bc} \\ -\frac{b}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix}$$

$$\therefore (\mathbf{A}^{-1})^T \equiv (\mathbf{A}^T)^{-1} \text{ for } 2 \times 2 \text{ matrices}$$